PROJECT

CET. CIBS. D.L.104

Cybersecurity Specialist Technician

UFCD 5892 – Network Management and Customer Support Models

Part 1

Addressing

LAN

Domain: cinel.pt

Net: 192.168.40.0/24

Pfsense IP: 192.168.40.254

DHCP range: 192.168.40.45 – 192.168.40.89

DNS IP: 192.168.40.1

Router / Firewall

OS: pfsense-CE-2.7.2

Hostname: pfsense

IP Interfaces: WAN (DHCP) | LAN (192.168.40.254)

Services: Certificate Authorship

The pfsense has to be by https and therefore a certificate must be created for this purpose and the port must be 7777.

pfsense must be accessible either by FQDN or by IP.

Active Directory / Domain controller

OS: Windows Server 2022 DataCenter Evaluation

Hostname: DC1

IP Interfaces: Ethernet0 (192.168.40.1)

Services: ADDC | DHCP | DNS

Description: As an ADDC server, you are responsible for all authentication of the cinel.pt domain for existing groups and users.

Kali Linux

Hostname: kali

OS: Kali Linux

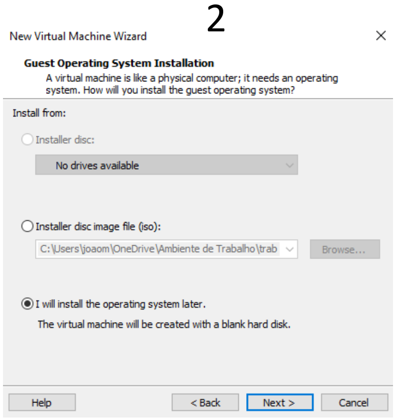
IP interfaces: Ethernet0 (DHCP)

Description: A MITM attack is to be made by capturing pfsense user credentials.

Router / Firewall Creation – PFSense

When opening VMware, I start by creating a new machine by following these steps:

1


2


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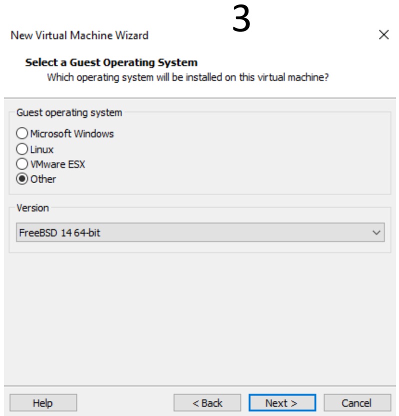

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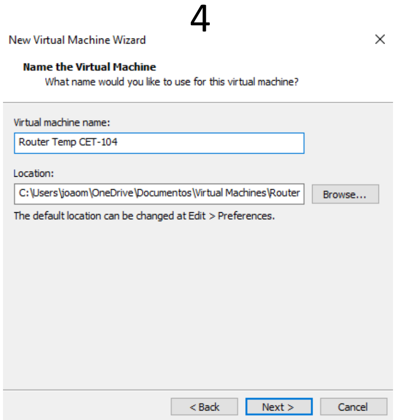

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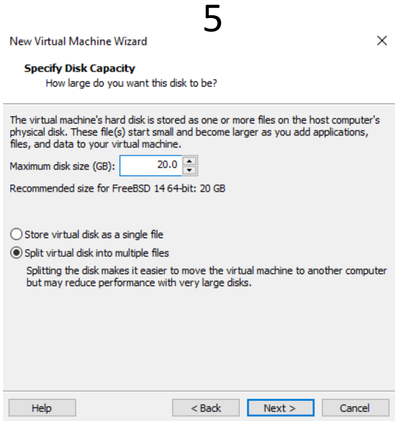

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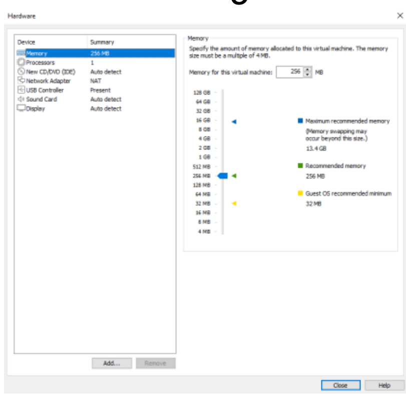

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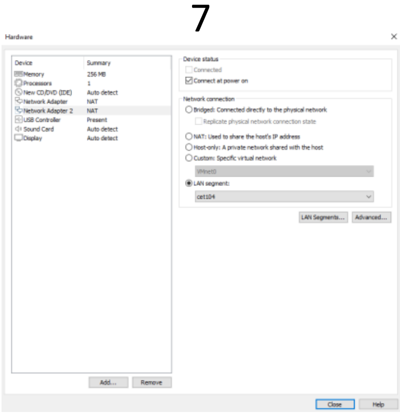

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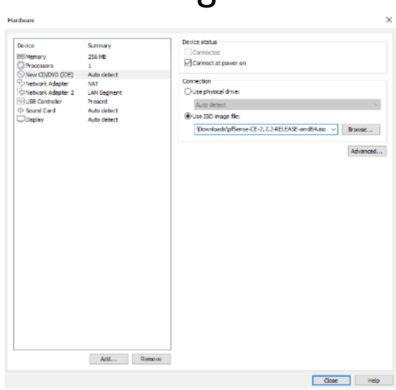

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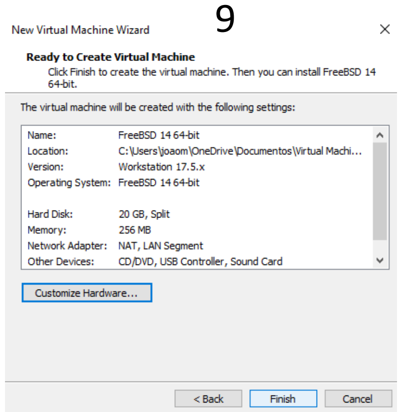

Next, I will install the operating system with these steps:

1

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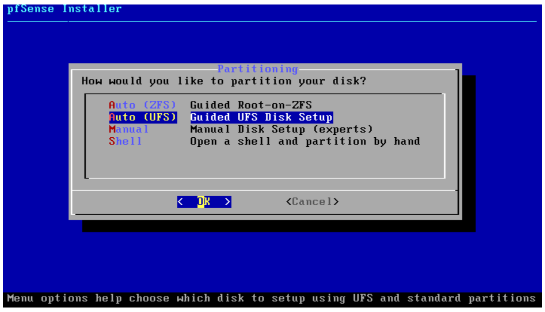

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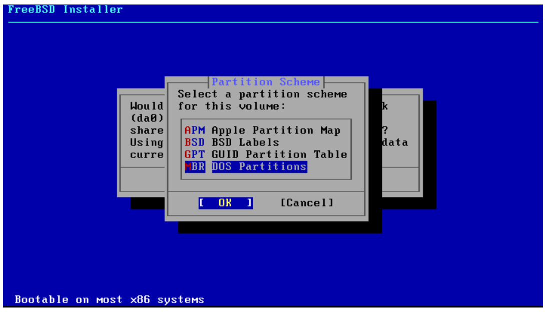

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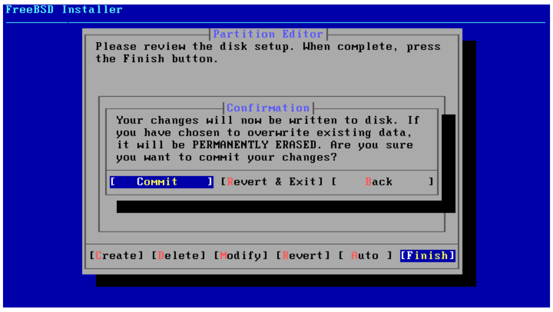


After rebooting you will be configured with these options:

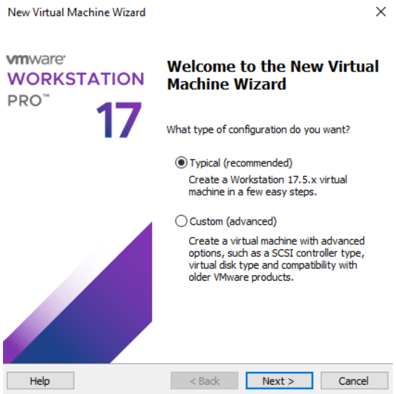
2) Interface IP Address: 2 → 2 → n → 192.168.40.254 → 24 → Enter → n → Enter → n → y → Enter

If necessary, you can also test the connection via the Shell (option 8) with the ping command.

Creation of Active Directory / Domain controller – Windows Server 2022

When opening VMware, I start by creating a new machine by following these steps:

1



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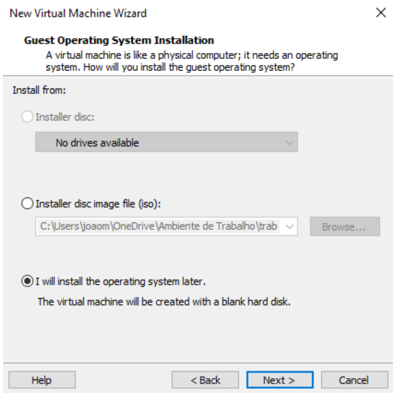

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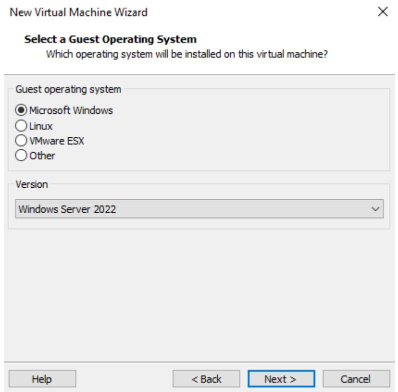

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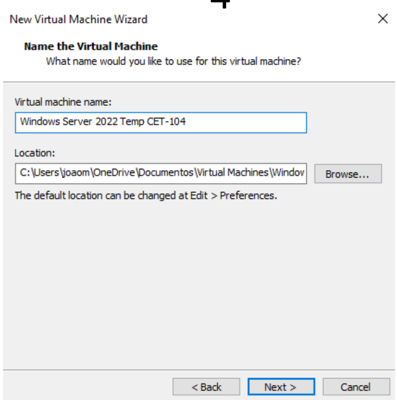

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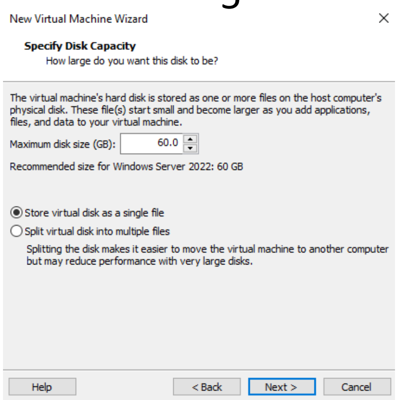

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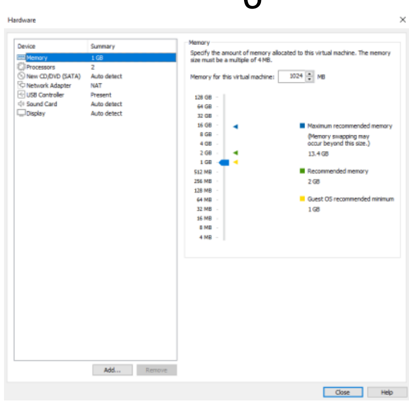

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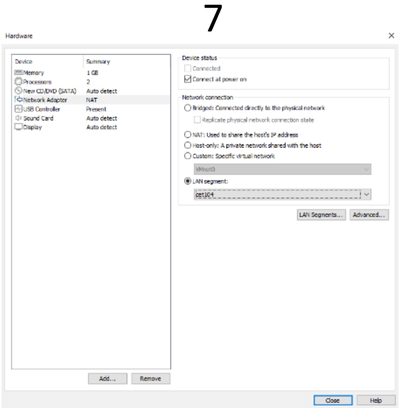

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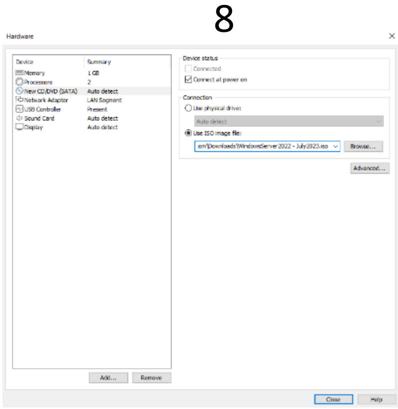

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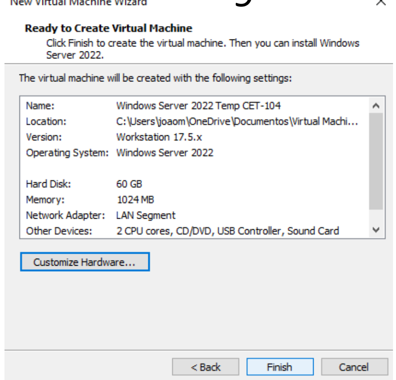
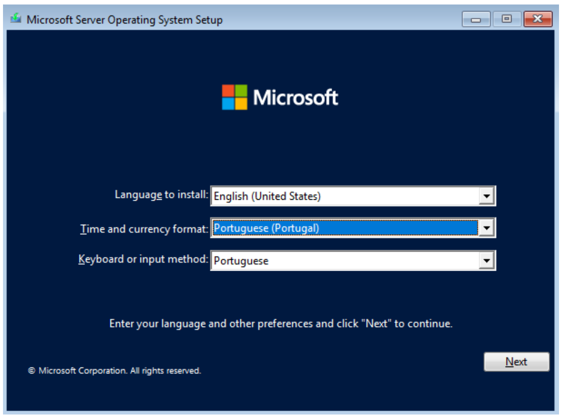


Operating System Installation – Windows Server 2022

Next, I will install the operating system with these steps:

1



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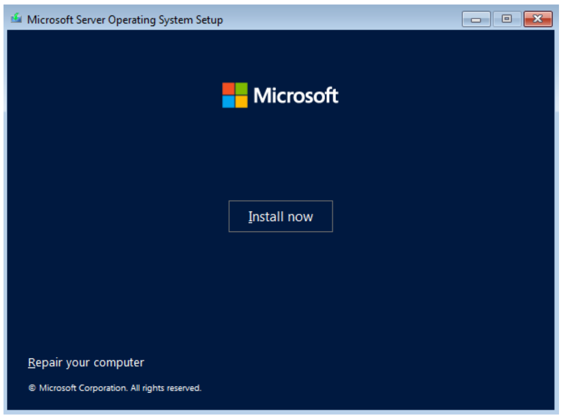

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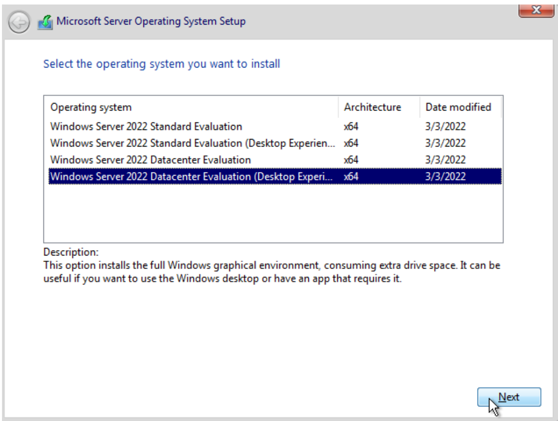

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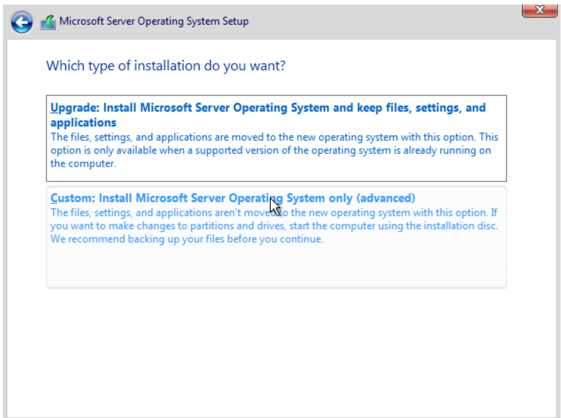

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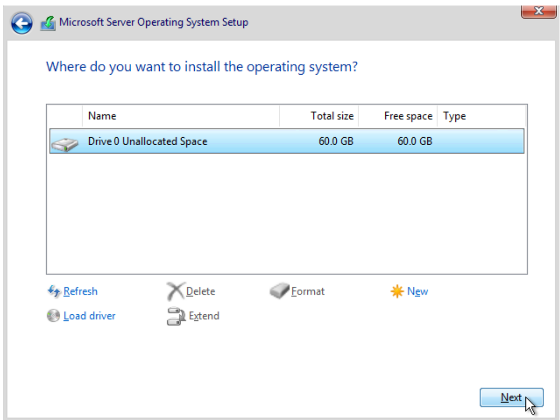
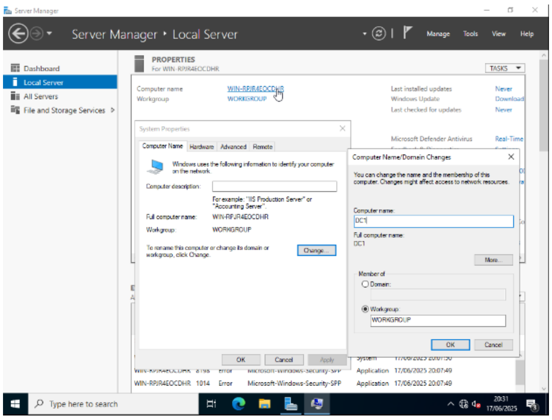
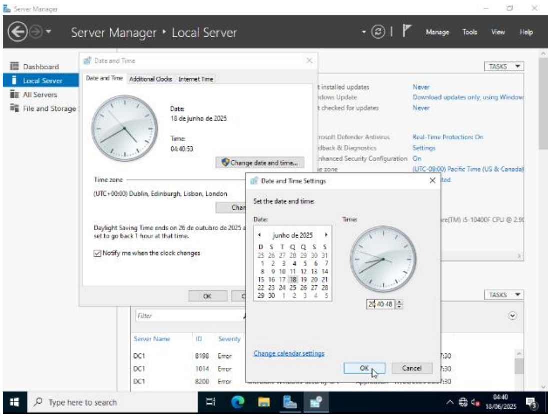
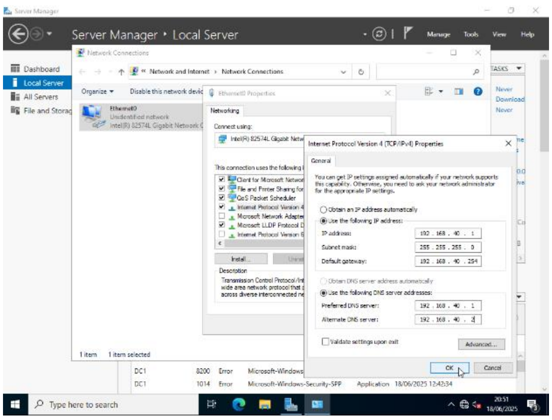


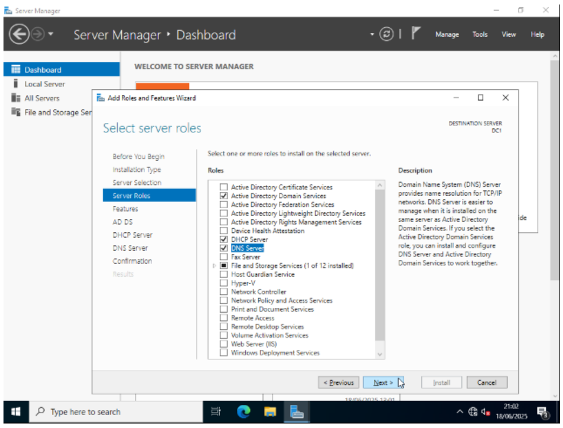
Windows Setup - Windows Server 2022

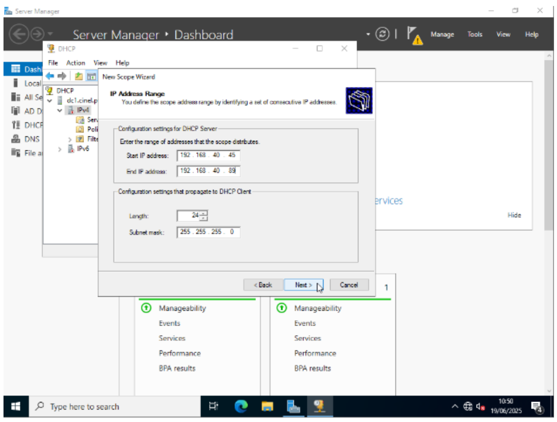
With the windows server installed I will make some final changes like:







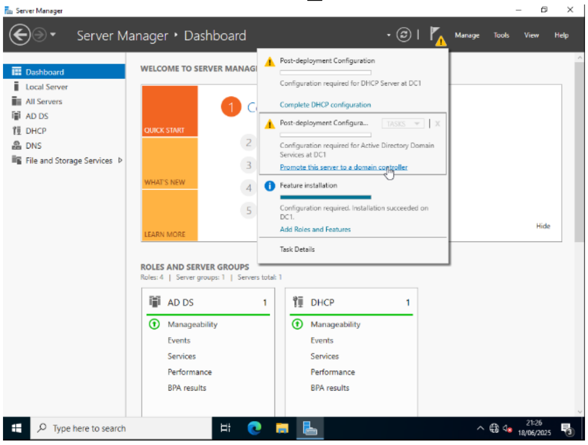




CINEL

Domain Configuration - Windows Server 2022

1



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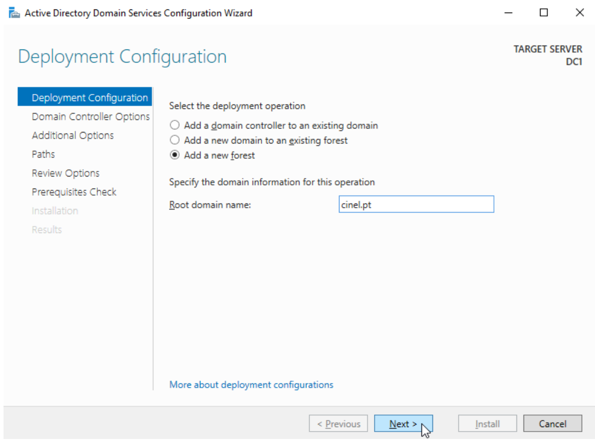

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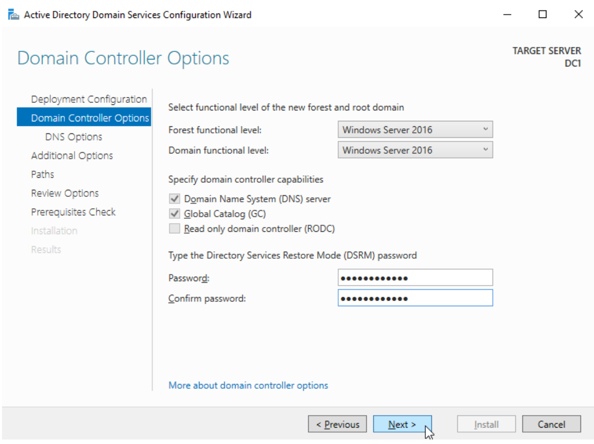

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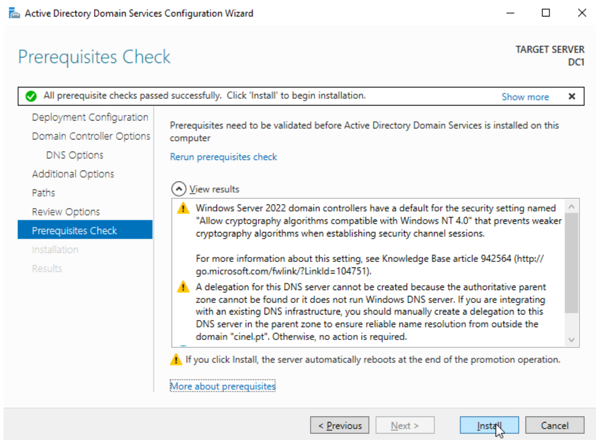


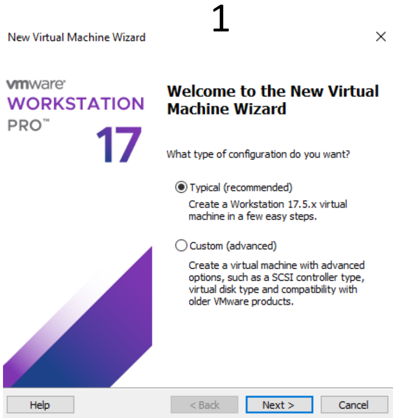
After making this configuration it is necessary to change the DNS Server Addresses" within the "Internet Protocol Version 4" as done before and the domain action resets it after it is implemented.

CINEL

Creation of Kali Linux

When opening VMware, I start by creating a new machine by following these steps:

1



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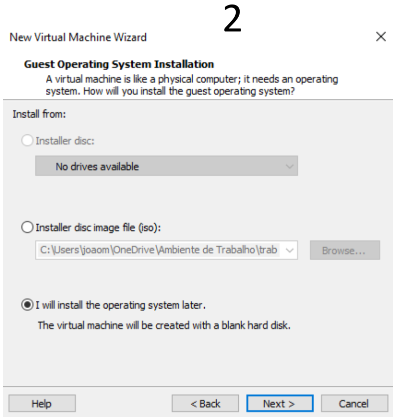

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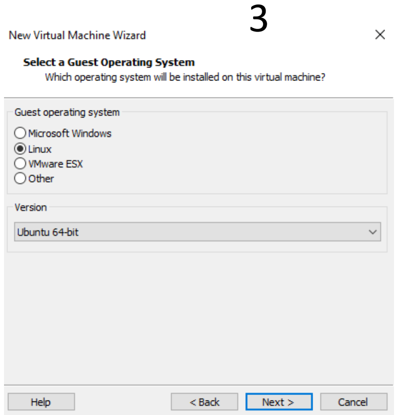

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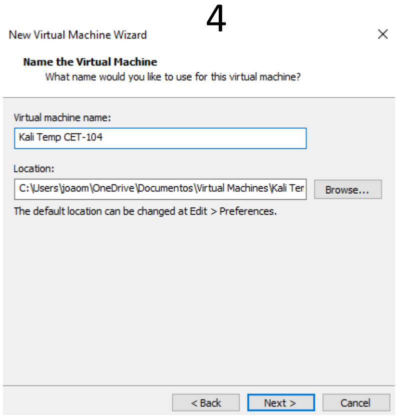

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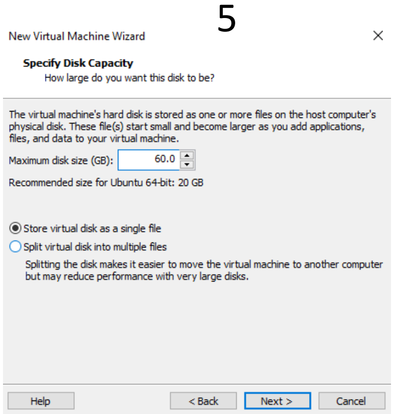

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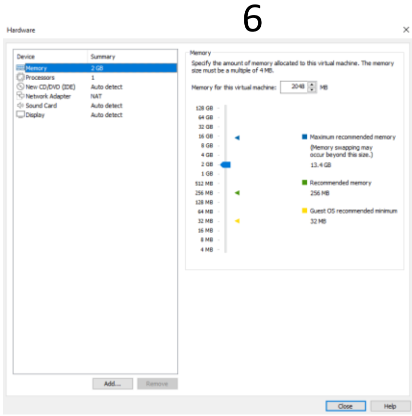

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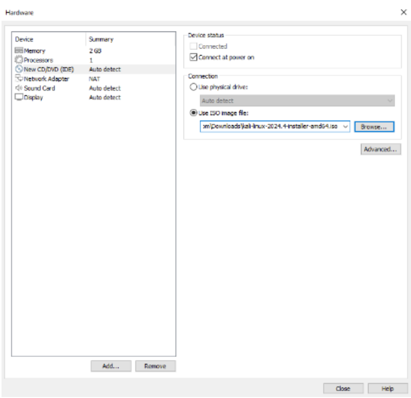

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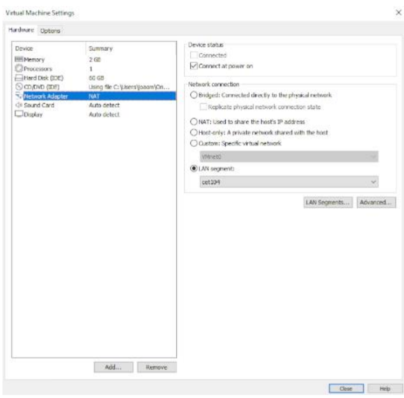

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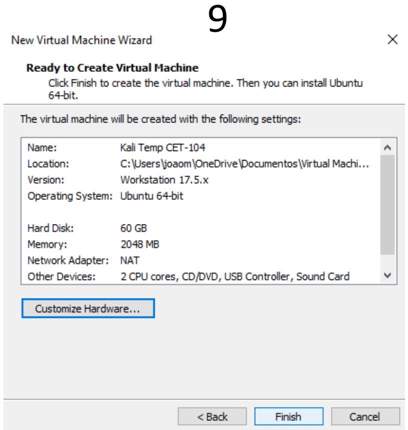

CINEL

Kali Linux Configuration

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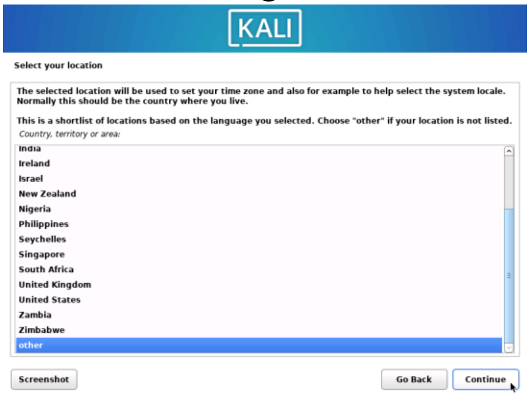

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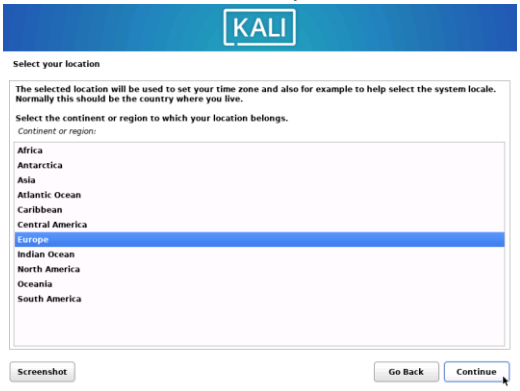

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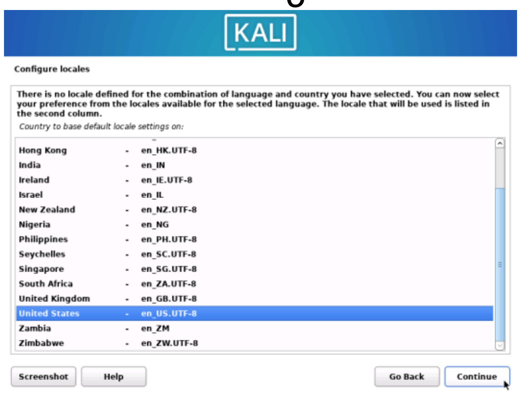

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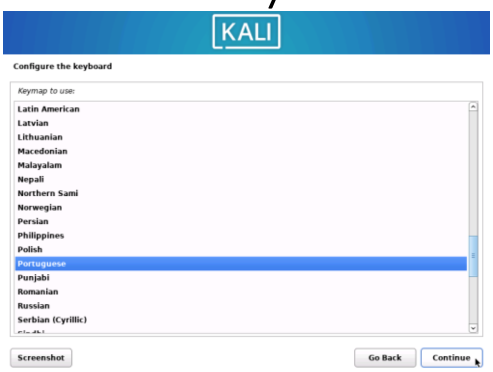

Kali Linux Configuration

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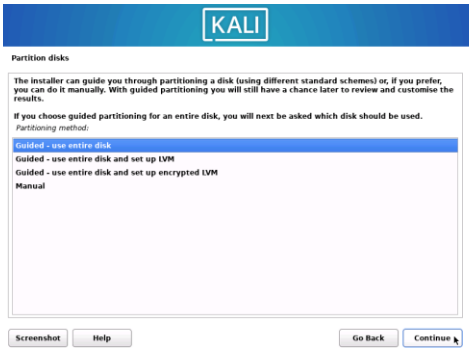

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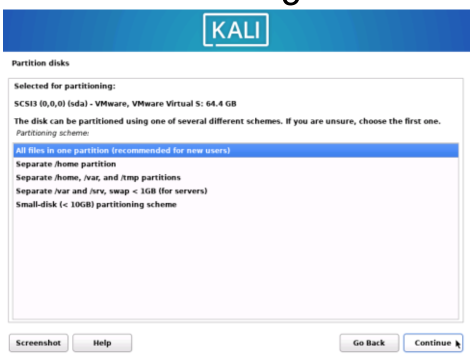

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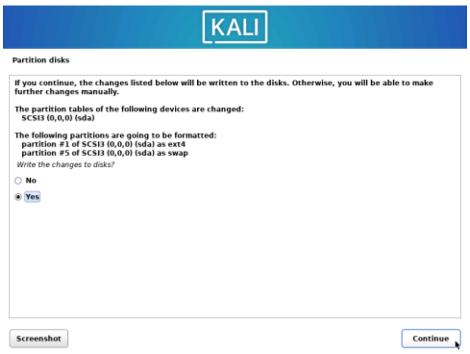

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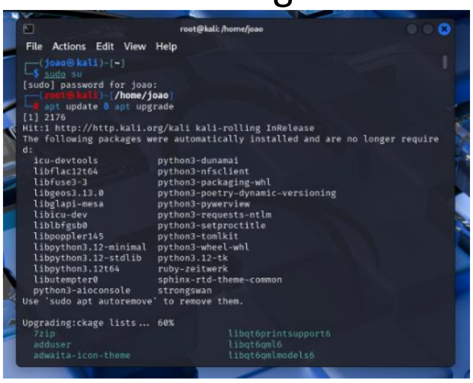
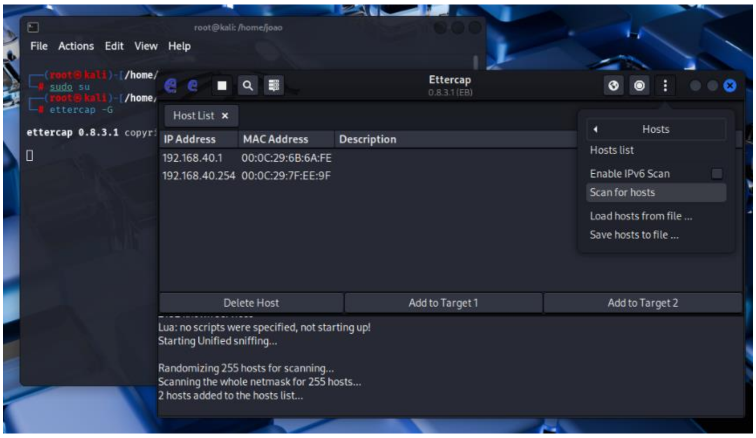


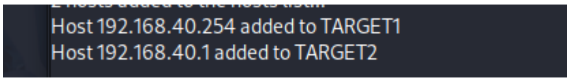
MITM Attack – Kali Linux

To get the user credentials from pfsense I will use the ettercap service available on kali Linux.

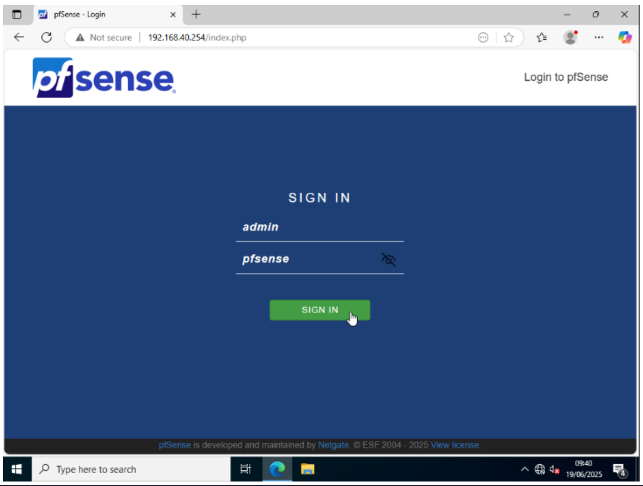
When opening ettercap, it first performs "Scan for hosts" to identify any IP present in the LAN segment.

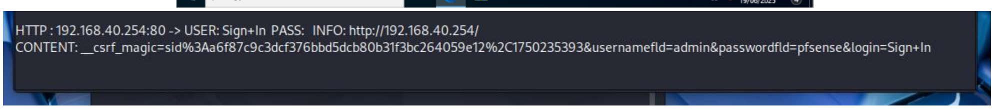


In this attack, router 192.168.40.254 will be "Target 1" and Windows server 192.168.40.1 will be "Target 2".



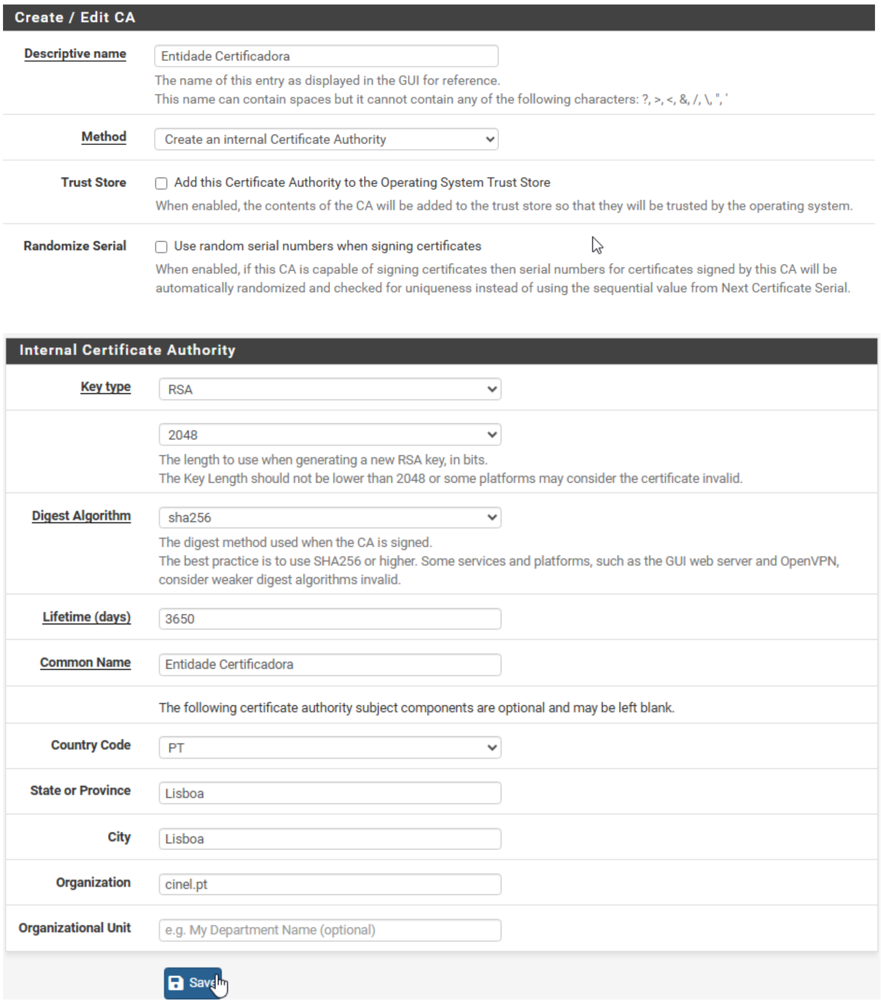
When entering an unencrypted http site with windows server, ettercap can obtain the windows server credentials by displaying the username and password used by the user.



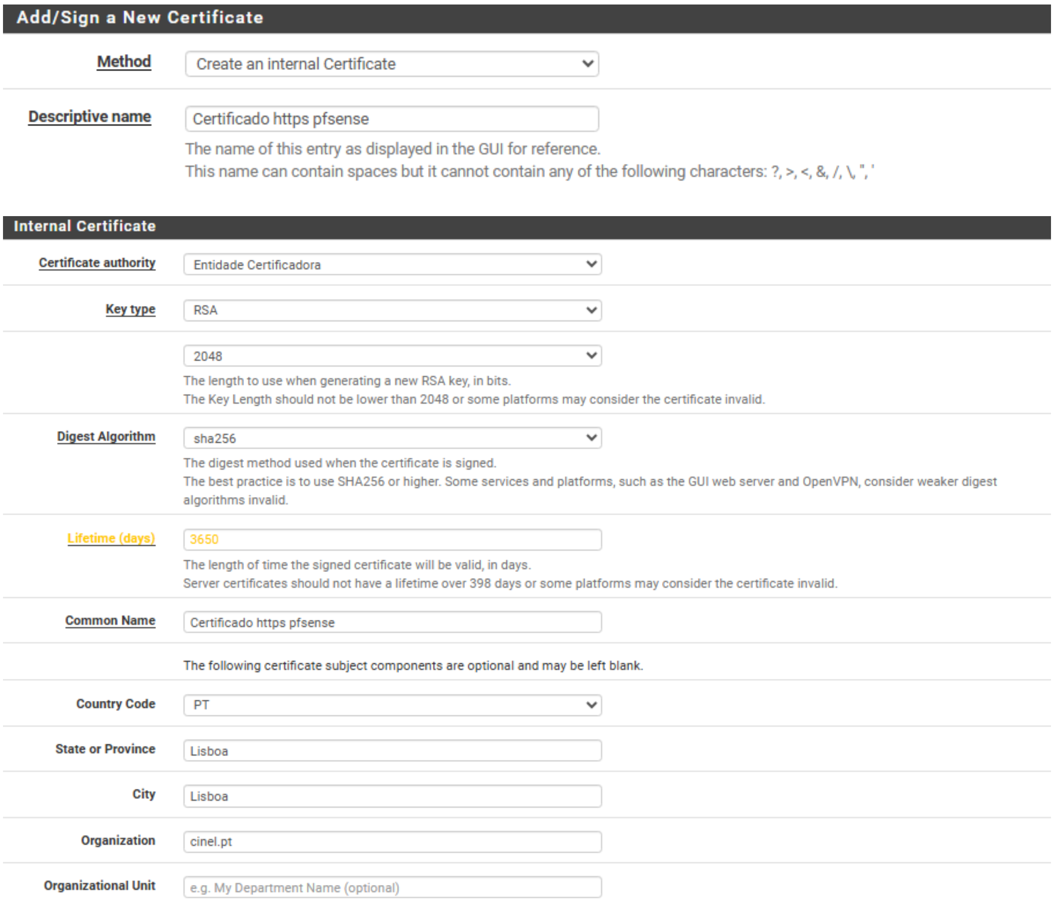


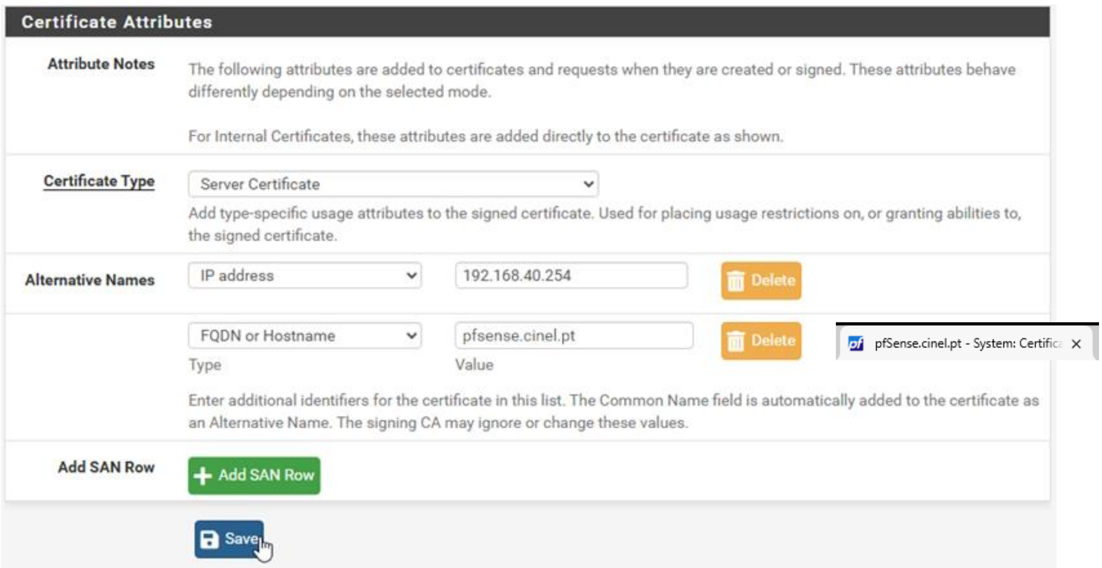
A defence technique against the attack previously made is the encryption of an address.

This can be done by creating a certifying body and a certificate on the http://192.168.40.254/ router, when logging in, go to System → Certificates → Authorities and Add and fill in the following information:

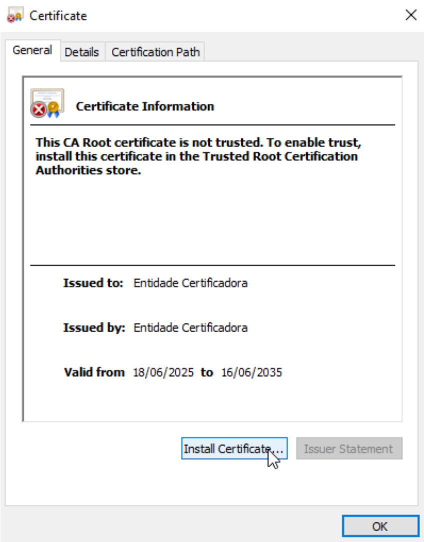


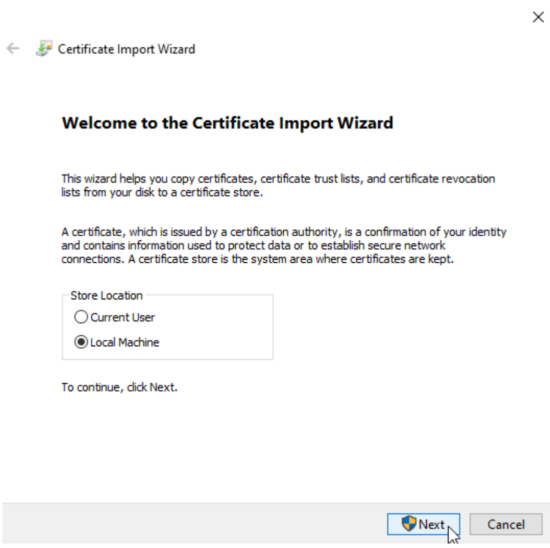
With the certifying body created, we will now create the certificate, for this go to System → Certificates → Certificates and Add/Sign by filling in the information:

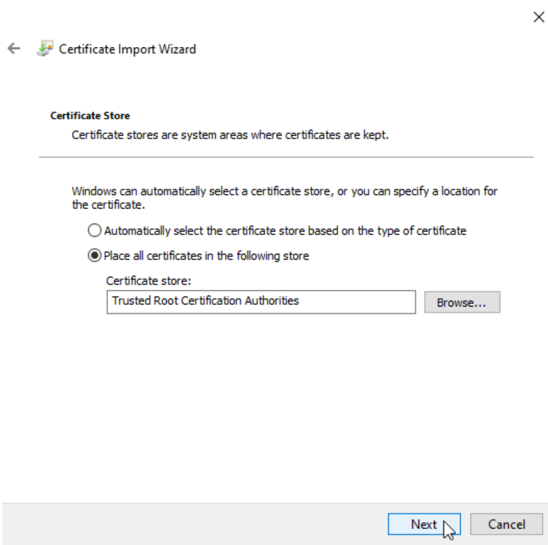




Then I'll extract both through the "Export" action and after downloading I'll open them and do this procedure for both:

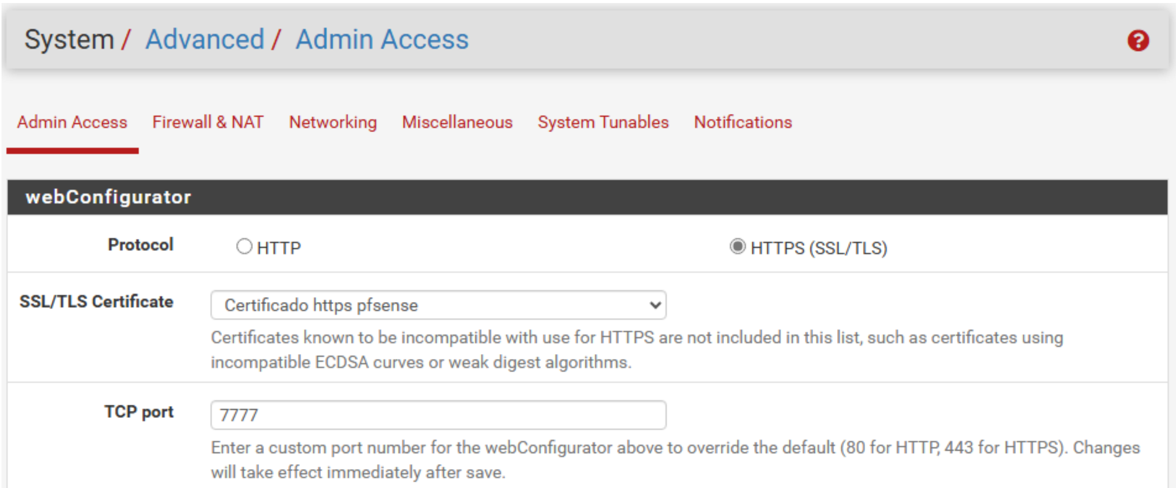




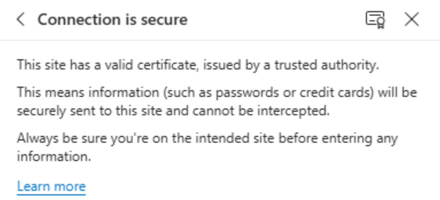


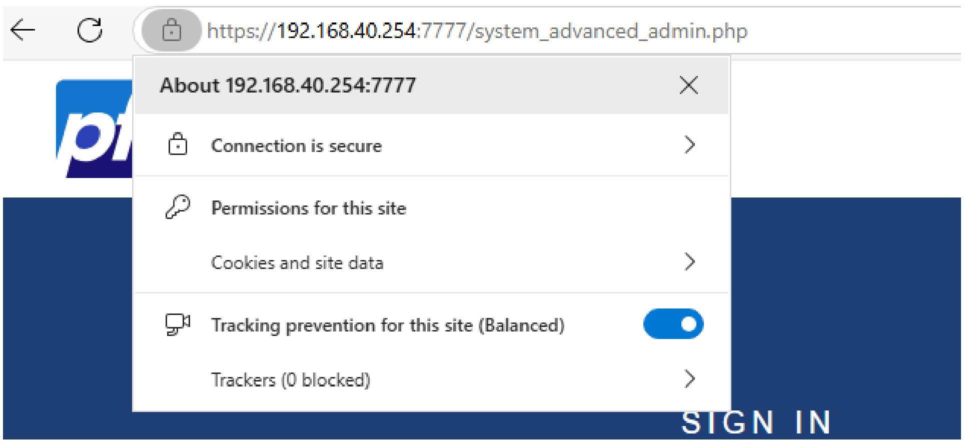
HTTPS Certified Defence – Windows Server 2022

Finally, to activate the https service, go to System → Advanced and fill in the updated information:



When you save, you will be redirected to the updated address with the https service enabled.





Menus that were not screenshotted in part 1 remained unchanged during the process, requiring only the action to advance to the next step.

Ex: "Next/Finish" confirmations, terms and conditions menus, Required restart notices.

Part 2

Task Summary (OSINT):

This is a photo of a resort located on an island. To perform this task, it is necessary for the trainees to explain how they arrived at the answers through print screens.

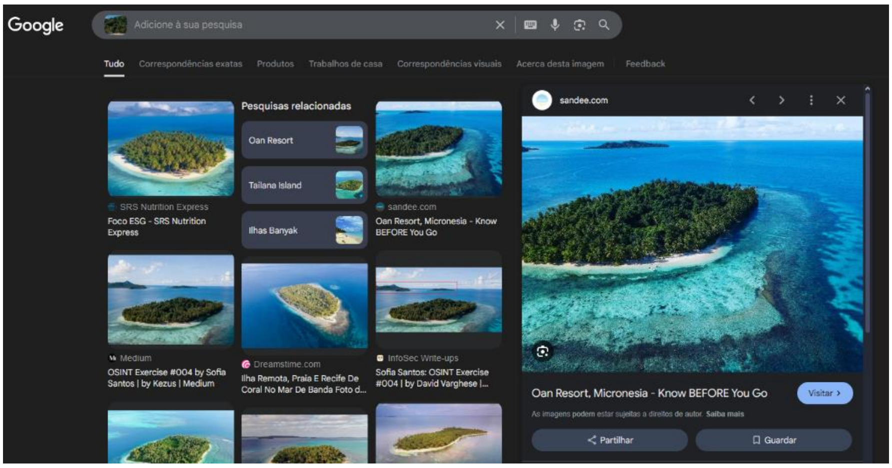
a) What is the name of the resort?

b) What are the coordinates of the island?

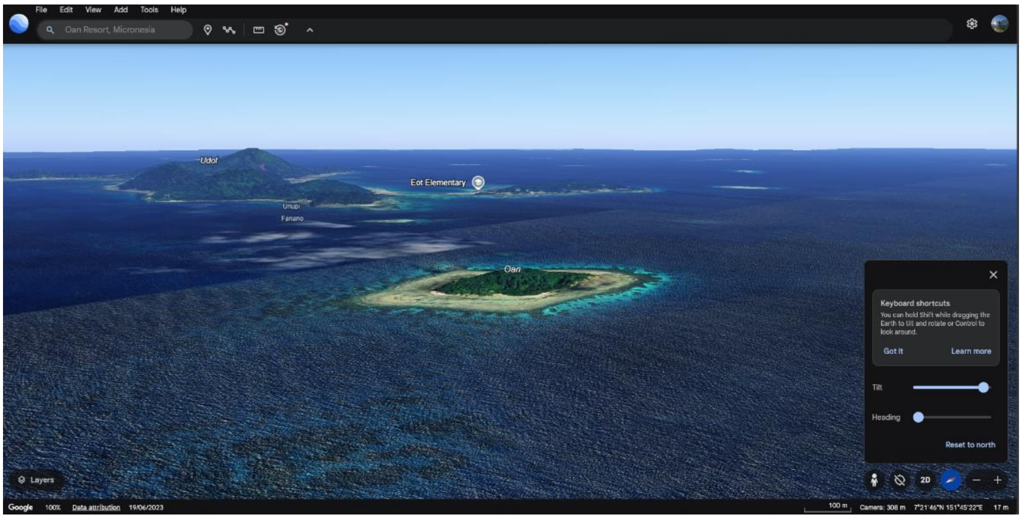
c) In which cardinal direction was the camera facing when the photo was taken?

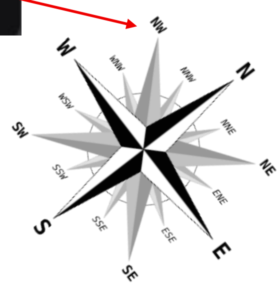
Part 2

To find out the exact location of the island I simply used reverse image search, eventually finding some results.



With the name of the island (Oan Resort) I can use google earth to see its coordinates and if I experiment with the camera options such as tilt, I get an angle similar to the image provided.





Coordinates: 7º21'46"N 151º45'22"E Cardinal Direction: NW

Part 3

Task Summary (OSINT)

This is a purposeful site for information gathering and hacking.

[Hack This Site]

According to what was done in class, it is requested that as much information as possible be collected about the target, which is the site above.

Everything they collect must be subject to proof, it can be through print screens. (IP's, Domains, Emails, Geolocation, etc.)

Part 3

To find the following information I used resources such as: maltego, dns dumpster, netcraft, wappalyzer, cmd (ping, nslookup), page source.

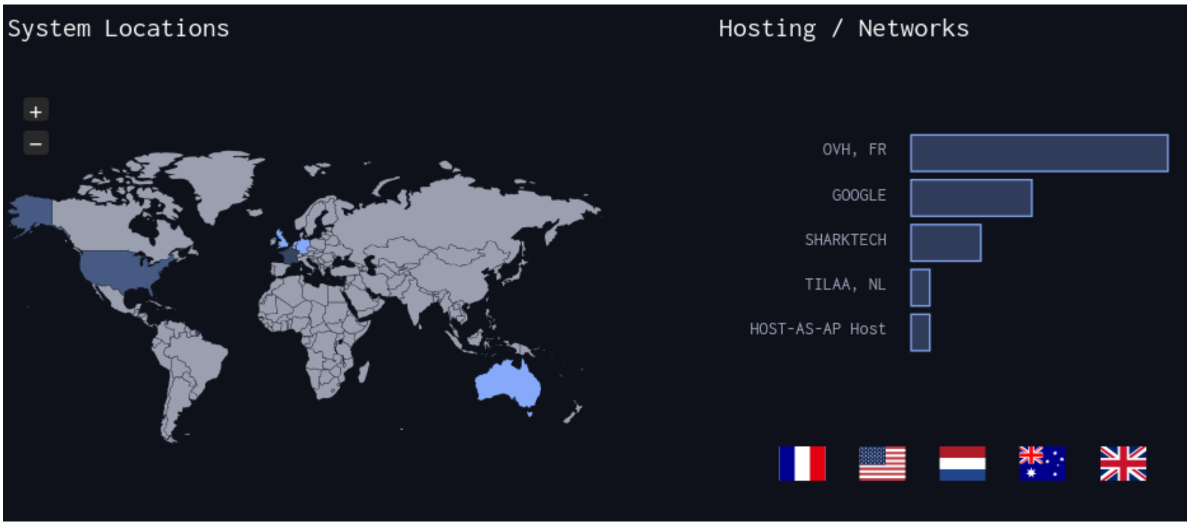
IP: 137.74.187.103

Server IP: 192.168.31.2

```
┌──(joao㉿kali)-[~]
└─$ ping www.hackthissite.org
PING www.hackthissite.org (137.74.187.103) 56(84) bytes of data.
```

```
┌──(joao㉿kali)-[~]
└─$ nslookup hackthissite.org
Server:         192.168.31.2
Address:        192.168.31.2#53

Non-authoritative answer:
Name:   hackthissite.org
Address: 137.74.187.103
Name:   hackthissite.org
Address: 137.74.187.100
Name:   hackthissite.org
Address: 137.74.187.101
Name:   hackthissite.org
Address: 137.74.187.104
Name:   hackthissite.org
Address: 137.74.187.102
```

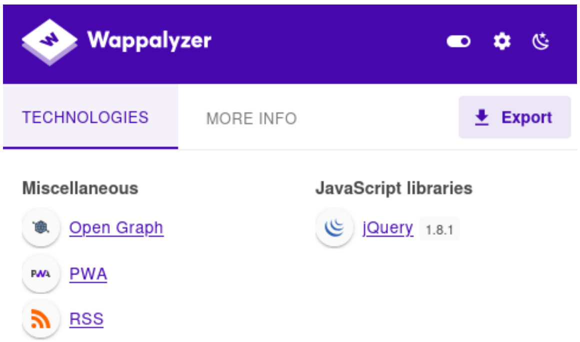


Languages / technologies: html, css, rss, xml, javascript, jquery, ssl, php, google webmaster tools, xhtml, open graph, pwa

```
<link href="https://data.htscdn.org/themes/Dark/Dark.css" rel="stylesheet" type="text/css" />
<link href="https://www.hackthissite.org/pages/hts.rss.php" rel="alternate" type="application/rss+xml" title="HTS RSS feed" />
<base href="https://www.hackthissite.org" />
<script type="text/javascript" src="https://data.htscdn.org/js/jquery-1.8.1.min.js"></script>
<script type="text/javascript" src="https://data.htscdn.org/js/notice.min.js"></script>
```

Network

Site	http://www.hackthissite.org	Domain	hackthissite.org
Netblock Owner	OVH Static IP	Nameserver	c.ns.buddyns.com
Hosting company	OVH	Domain registrar	pir.org
Hosting country	NL	Nameserver organisation	whois.domrobot.com
IPv4 address	137.74.187.101 (VirusTotal)	Organisation	Data Protected, REDACTED, REDACTED, REDACTED, US
IPv4 autonomous systems	AS16276	DNS admin	admin@hackthissite.org
IPv6 address	Not Present	Top Level Domain	Organization entities (.org)
IPv6 autonomous systems	Not Present	DNS Security Extensions	Enabled
Reverse DNS	hackthissite.org		



Some Sub-Domains:

Host	IP	ASN	ASN Name	Open Services (from DB)	RevIP
api.hackthissite.org	137.74.187.116 api.hackthissite.org	ASN: 16276 137.74.0.0/16	OVH, FR France		1
vm-005.outbound.firewall.hackthissite.org	137.74.187.154 vm-005.outbound.firewall.hackthissite.org	ASN: 16276 137.74.0.0/16	OVH, FR France		2
vm-050.outbound.firewall.hackthissite.org	137.74.187.155 vm-050.outbound.firewall.hackthissite.org	ASN: 16276 137.74.0.0/16	OVH, FR France		2
vm-099.outbound.firewall.hackthissite.org	137.74.187.159 vm-099.outbound.firewall.hackthissite.org	ASN: 16276 137.74.0.0/16	OVH, FR France		2
vm-150.outbound.firewall.hackthissite.org	137.74.187.158 vm-150.outbound.firewall.hackthissite.org	ASN: 16276 137.74.0.0/16	OVH, FR France		2
vm-200.outbound.firewall.hackthissite.org	137.74.187.157 vm-200.outbound.firewall.hackthissite.org	ASN: 16276 137.74.0.0/16	OVH, FR France		2
git.hackthissite.org	137.74.187.145 git.hackthissite.org	ASN: 16276 137.74.0.0/16	OVH, FR France		2
irc.hackthissite.org	137.74.187.150 lille.irc.hackthissite.org	ASN: 16276 137.74.0.0/16	OVH, FR France	https: unknown server title: HackThisSite IRC cn: www.irc.hackthissite.org	2
irc-hub.hackthissite.org	198.148.81.169 customer.sharktech.net	ASN: 46844 198.148.81.0/24	SHARKTECH United States		1
lille.irc.hackthissite.org	137.74.187.150 lille.irc.hackthissite.org	ASN: 16276 137.74.0.0/16	OVH, FR France	https: unknown server title: HackThisSite IRC cn: www.irc.hackthissite.org	2
wolf.irc.hackthissite.org	185.24.222.13 wolf.irc.hackthissite.org	ASN: 196752 185.24.220.0/22	TILAA, NL The Netherlands	http: nginx/1.18.0 (Ubuntu) title: 301 Moved Permanently https: nginx/1.18.0 (Ubuntu) title: 301 Moved Permanently cn: www.irc.hackthissite.org	2

Discovered through Maltego: ips, emails, different domains, locations, dns names, documents, records

IPS: 137.74.187.103, 137.74.187.97, 137.74.187.98, 137.74.187.96

Emails: admin@hackthissite.org, staff@hackthissite.org, spaux@hackthissite.org, kage@hackthissite.org, mail.patriciavaughn020@gmail.com, 88966567@qq.com, weekend@hackthissite.org, boschhacklord@gmail.com, domainabuse@ducows.com, 363081675@qq.com,andrewallanhacks@gmail.com, hackethics008@gmail.com, hack.consultant@hackermail.com, sulthu1274@gmail.com

Locations: Netherlands (hosting country), Admin Street, Province, Admin State/Province